

Time-frequency decomposition-based weighted ensemble learning for motor imagery EEG classification

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Abstract—Motor imagery brain-computer interface system based on Electroencephalogram (EEG) is an effective way to help the disabled recover part of their motor abilities. However, decoding the movement intention contained in the EEG signal accurately presents many challenges. In this paper, we propose a time-frequency decomposition-based weighted ensemble learning (TFDWEL) method, which aims to improve the classification performance of motor imagery EEG signals. The TFDWEL method divides the EEG signal into multiple subsets, and uses four time-frequency processing methods to extract the time-frequency sub-bands of each subset. Then the feature extraction model and classifier model of each subset trained by the common spatial pattern (CSP) algorithm and the support vector machine (SVM) algorithm are used to build a set of base learners. The least square error estimation method is used to learn the weight of each base learner, and finally the weighted summation method is used to obtain the final decision. The classification performance of the TFDWEL method is evaluated on the BCI Competition IV Data Set 2b, and the results show that the classification accuracy of 81.58% can be obtained. Superior classification performance indicates that the TFDWEL method can be used in further research to help the rehabilitation of the disabled.

I. INTRODUCTION

Brain-computer interface (BCI) system provides a new mode of communication for the interaction between humans and external devices. It does not need to establish a pathway through peripheral nerves, and directly transmits the deep-level brain movement consciousness to external control devices [1]. For those who suffer from neuromuscular disorders, such as paralysis, amyotrophic lateral sclerosis (ALS), spinal cord injury (SCI), the BCI system is a very effective solution, allowing patients to recover their daily exercise ability with the help of external equipment such as exoskeleton, prostheses, and orthotics [2].

Electroencephalogram (EEG) is a non-invasive method with great flexibility and security, and has attracted great attention in the research of BCI technology [3]. When the brain imagines the movement of the limbs on the body, the electric

potentials in the sensorimotor area of the cerebral cortex will produce rhythmic activity. This phenomenon is called event-related desynchronization (ERD) and event-related synchronization (ERS). Therefore, decoding the sensorimotor rhythm (SMR) of the motor cortex during motor imagery (MI) can obtain the brain's motor intention information in advance, which makes SMR become the commonly used analysis potential in EEG [4].

Decoding the motor intention in MI EEG signals presents many challenges [5]. The EEG signal recorded from the cerebral cortex is a time series of multi-channel potential activity, and the signal amplitude is very weak, which makes the EEG signal low signal-to-noise ratio and non-stationary [6]. In addition, due to the individual differences of the subjects, the response delay time of each subject to the task cue is different, and a fixed time window of the EEG signal may lead to poor classification performance [7].

Many researchers devoted to developing algorithms to improve the classification performance of motor imagery EEG signals [8]–[10]. Common spatial pattern (CSP) is the most commonly used algorithm for extracting discriminative EEG features related to MI [9]. However, the performance of CSP is greatly affected by the frequency band of the EEG signal, and the CSP features extracted from a single frequency band cannot fully represent the information of the raw EEG signal. The proposed improved versions of CSP, such as CSSP, RCSP, FBCSP [10], have improved the performance of CSP by adding information in the temporal domain and spectral domain.

Support vector machine is one of the most widely used classifiers in the EEG signals [11]. It achieves good generalization performance by maximizing the boundary distance between classes, and improves the classification results of MI EEG signals.

It has been well demonstrated that ensemble learning combines multiple base learners to achieve unified decision-making of multiple classifiers, and shows better classification performance than that using a single classifier [11] [12]. Bagging algorithm is one of the most popular algorithms in ensemble learning [13]. It builds and trains a set of base learners by sampling different data subsets, and then merges the decisions of all base learners, which can reduce the complexity of the model, and be robust to the variance of EEG signal both in the temporal domain and inter-subject [14].

In this paper, we propose a novel algorithm, namely time-frequency decomposition-based weighted ensemble learning (TFDWEL). This method divides the EEG signal into

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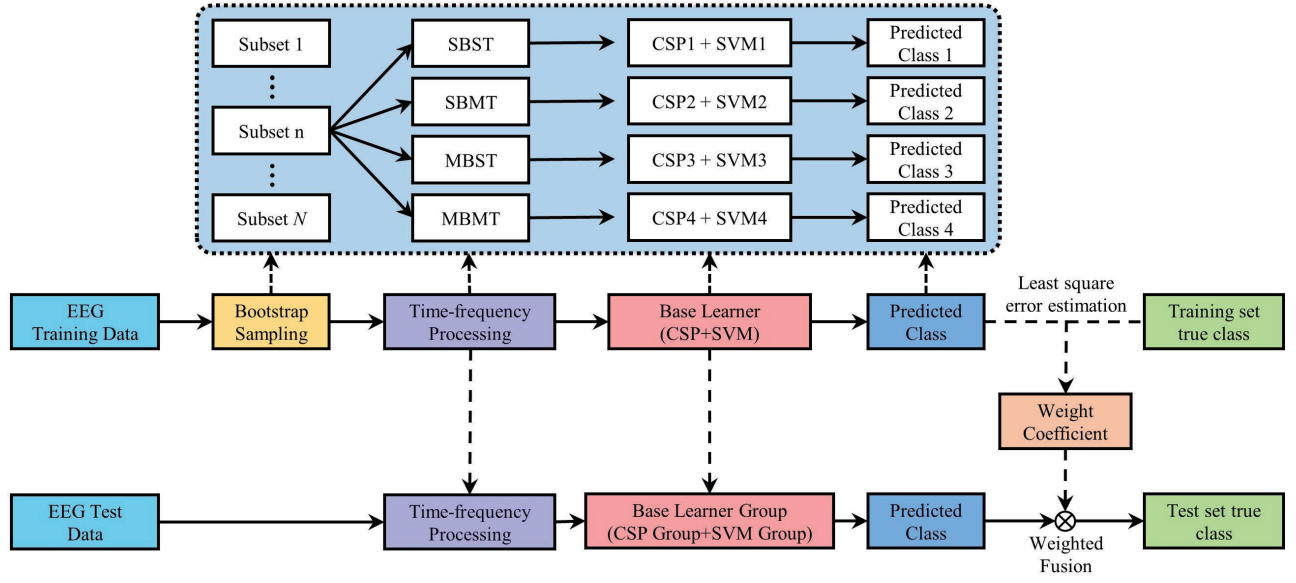


Fig. 1. Overall framework of the proposed TFDWEL method for MI EEG classification.

multiple subsets by bootstrap sampling, and then uses four different time-frequency processing methods to extract the time-frequency sub-bands of each subset to obtain subsets with different time-frequency distributions. We train the feature extraction model and classifier model of each subset through CSP algorithm and SVM algorithm, and then construct a set of base learners from all feature extraction models and classifier models. The least square error estimation method is used to learn the weight of each base classifier, and the weighted summation method is used to obtain the final decision. We evaluate the performance of the proposed method on the commonly used BCI Competition IV Data Set 2b. Experimental results show that the proposed TFDWEL method can effectively distinguish the class of MI EEG signal, with the highest classification accuracy reaching 81.58%.

The organization of this paper is as follows: the TFDWEL method is described in Section II. The EEG dataset and experimental protocol are introduced in Section III. The experimental results are presented in Section IV. Conclusions is given in Section V.

II. METHODOLOGY

In this section, we describe the proposed time-frequency decomposition-based weighted ensemble learning method in detail, and the classification framework of this method is schematically depicted in Fig. 1.

A. Data sampling

One of the most typical representatives of parallel ensemble learning algorithms is Bagging. The algorithm samples the training data set through the bootstrap sampling method to generate several different subsets, and then trains a base learner from each subset, and then summarizes the predicted classes of all the base learners to make the final decision.

Supposing the given training set $X_{train} \in \mathbb{R}^{p \times c \times tr}$, where c is the number of channels, p is the number of temporal

points in each trial, and tr is the number of trials. We use the strategy of bootstrap sampling method to sample N training subsets of equal size from the training set X_{train} , and use the method of sampling with replacement to get k samples of each subset, and then the new training set $S = [s_1, s_2, \dots, s_N] \in \mathbb{R}^{p \times c \times k}$.

B. Time-frequency processing

In the time-frequency processing stage, we design four processing methods: single-band-single-time (SBST), single-band-multi-time (SBMT), multi-band-single-time (MBST) and multi-band-multi-time (MBMT). In single-band, a zero-phase Butterworth filter performs band-pass filtering over a wide frequency band. In multi-band, we divide the frequency band of the response signal of brain motor imagery into F overlapping sub-bands, and then use a zero-phase Butterworth filter to perform band-pass filtering on each overlapping sub-band. In single-time, according to the time schedule of the MI task in the experimental paradigm, temporal points in a specific time window after filtering are extracted. In multi-time, the time period of imaging movement in the experimental paradigm is divided into T overlapping time windows, and then the temporal points in each time window are extracted respectively.

Applying the four time-frequency processing methods to the N subsets obtained by the bootstrap sampling method, we can get $4N$ subsets with time-frequency differences, as follows:

$$TF = \begin{bmatrix} tf_{1,1} & \dots & tf_{1,N} \\ \vdots & \ddots & \vdots \\ tf_{h,1} & \dots & tf_{h,N} \end{bmatrix}, \quad (1)$$

where $h = [SBST, SBMT, MBST, MBMT]$, and for each time-frequency sub-band, T represents the time window, F represents the frequency band, as follows:

$$tf_{i,j} = \begin{bmatrix} t_{1,1} & \dots & t_{1,F} \\ \vdots & \ddots & \vdots \\ t_{T,1} & \dots & t_{T,F} \end{bmatrix}. \quad (2)$$

C. Feature extraction via CSP

Common spatial pattern (CSP) is the most commonly used feature extraction algorithm in MI-BCIs. The algorithm uses the method of supervised learning and the diagonalization of the matrix to find an optimal spatial filter in the binary EEG signal, making the difference between the two classes of signals the largest.

For the time-frequency processed signal TF , the normalized covariance matrices of the two classes are R_1 and R_2 respectively, as follows:

$$R_1 = \frac{TF_1 TF_1^\top}{\text{trace}(TF_1 TF_1^\top)}, R_2 = \frac{TF_2 TF_2^\top}{\text{trace}(TF_2 TF_2^\top)}, \quad (3)$$

where $\top$ represents the transposition of the matrix, and $\text{trace}(\cdot)$ is the sum of the elements on the diagonal of the matrix.

Calculate the mixed covariance matrix R :

$$R = \overline{R_1} + \overline{R_2}, \quad (4)$$

where $\overline{R_1}$, $\overline{R_2}$ represents the average covariance matrix of class 1 and 2, respectively.

Then we calculate the eigenvalue decomposition of the mixed covariance matrix to obtain the whitening matrix P :

$$P = \sqrt{\lambda^{-1}} U^\top \quad (5)$$

where λ is the diagonal matrix formed by the corresponding eigenvalues in descending order, and U is the eigenvector matrix formed by the matrix λ .

We calculate the whitening transformation of the covariance matrix:

$$R'_1 = P_1 \overline{R_1} P_1^\top, R'_2 = P_2 \overline{R_2} P_2^\top, \quad (6)$$

We calculate the principal component decomposition of R'_1 and R'_2 :

$$R'_1 = B_1 \lambda_1 B_1^\top, R'_2 = B_2 \lambda_2 B_2^\top, \quad (7)$$

At the same time, the sum of the diagonal matrix λ_1 and λ_2 should be the identity matrix I , as follows:

$$\lambda_1 + \lambda_2 = I. \quad (8)$$

We extract the eigenvectors corresponding to the M largest and smallest eigenvalues to obtain the spatial filter w :

$$w = B^\top P. \quad (9)$$

Finally, we calculate the CSP feature set Z :

$$Z = w \times TF = \begin{bmatrix} Z_{1,1} & \dots & t_{1,N} \\ \vdots & \ddots & \vdots \\ Z_{h,1} & \dots & t_{h,N} \end{bmatrix}, \quad (10)$$

where $z_{i,j} \in \mathbb{R}^{k \times 2M \times T \times F}$ is the CSP features obtained from the j -th training subset under the i -th time-frequency processing method.

D. Feature classification

Support vector machine (SVM) classifier has become the most commonly used classifier in MI EEG signal classification due to its excellent learning ability [11]. We use the features obtained from the $4N$ training subsets to train the SVM classifier group, and then apply the test data to the classifier group to obtain the predicted class P_{test} of the test set:

$$P_{test} = \begin{bmatrix} ptest_{1,1} & \dots & ptest_{1,4N} \\ \vdots & \ddots & \vdots \\ ptest_{te,1} & \dots & ptest_{te,4N} \end{bmatrix}, \quad (11)$$

where $p_{test} \in [-1, 1]$, and te is the number of trials in the test set.

E. Fusion strategy

One of the most important factors affecting the performance of ensemble learning is the fusion strategy that incorporates the decisions of each classifier [15]. We use a weighted summation fusion method to integrate the decisions of each classifier, as follows:

$$T_{test} = \text{sign}\left(\sum_{k=1}^{te} P_{test} \times \beta\right), \quad (12)$$

where β is the weight of the classifier, and T_{test} is the final predicted class of test set.

The weight of the classifier is obtained by the least square error estimation algorithm. We use the classifier group obtained in the classification stage to predict the class labels of the $4N$ training subset features respectively, and then connect all the predicted labels. The prediction class P_{train} of the training set is as follows:

$$P_{train} = \begin{bmatrix} ptrain_{1,1} & \dots & ptrain_{1,4N} \\ \vdots & \ddots & \vdots \\ ptrain_{k \times 16N^2, 1} & \dots & ptrain_{k \times 16N^2, 4N} \end{bmatrix}, \quad (13)$$

Assuming that the sample class predicted by the classifier group has a multiple linear relationship with the true class, the multiple linear model is defined as follows:

$$T_{train} = P_{train} \beta + \varepsilon, \quad (14)$$

where $T_{train} \in \mathbb{R}^{k \times 16N^2 \times 1}$ is the true class of each trial in the training subset, $\beta \in \mathbb{R}^{4N \times 1}$ is the weight coefficient of the classifier, and $\varepsilon \in \mathbb{R}^{n \times 1}$ is a random variable with a mean value of zero.

We calculate the distance between the true class T_{train} and the predicted class $T_{train} \beta$, and the distance $Q(\beta)$ is defined as follows:

$$Q(\beta) = (T_{train} - P_{train} \beta)^\top (T_{train} - P_{train} \beta), \quad (15)$$

then calculate the partial derivative of β and set $Q(\beta)' = 0$, we can obtain as follows:

$$P_{train}^\top P_{train} \beta = P_{train}^\top T_{train}, \quad (16)$$

then the weight of each classifier is β :

$$\beta = (P_{train}^\top P_{train})^{-1} P_{train}^\top T_{train}. \quad (17)$$

where $[\cdot]^{-1}$ represents the inverse of matrix.

III. EXPERIMENTAL STUDY

A. EEG Data Description

The BCI Competition IV dataset 2b is used to evaluate the performance of our proposed method. This dataset contains the EEG of nine healthy subjects, which were recorded from left hand and right-hand motor imagery task. The electrodes C3, C4 and Cz were recorded with a sampling frequency of 250Hz. In this work, only the third training sessions was considered for analysis, i.e., 'B0103T', 'B0203T', ..., 'B0903T'. For each subject, a total of 160 trials (80 for each MI task) are available, and the MI task lasts 4.5s for each trial. For more details about this dataset, please refer to the website <http://www.bbc.de/competition/iv/>.

B. Experimental protocol

The purpose of the experiment is to analyze and evaluate the classification performance of our proposed TFDWEL method. In order to evaluate the classification performance more accurately, we use 10×10 -fold cross-validation to evaluate the TFDWEL method. For each subject, we randomly divide the trials of performing the left-hand and right-hand MI tasks into ten equal parts, one for the test set, and the remaining data for the training set, and the average result of 10 folds is the classification accuracy of a single experiment.

In the time-frequency processing stage, the filter passband of the single-band method is 8-35Hz, and the multi-band method divides the frequency band between 4Hz and 40Hz into 17 overlapping sub-bands, (i.e., 4-8Hz, 6-10Hz, ..., 36-40Hz), and performs band-pass filtering on each sub-band. The single-time method extracts the temporal points in a single time window (0.5-3.5s) after the appearance of the MI task cue from the filtered data. The multi-time method divides the 4s time window after the appearance of the MI task cue into 5 overlapping time windows, (i.e., 0-2s, 0.5-2.5s, ..., 2-4s), and extracts the temporal points in each time window from the filtered data. Therefore, after time-frequency processing, we can get $4 \times N$ different training subsets.

In the feature extraction stage, the CSP algorithm trains all the training subsets to obtain the CSP feature extraction model respectively. The filter pairs of the CSP algorithm is 1, so the feature numbers of the subsets processed by the SBST, SBMT, MBST, and MBMT methods after feature extraction are $2M \times T \times F = 2, 10, 34, 170$, respectively.

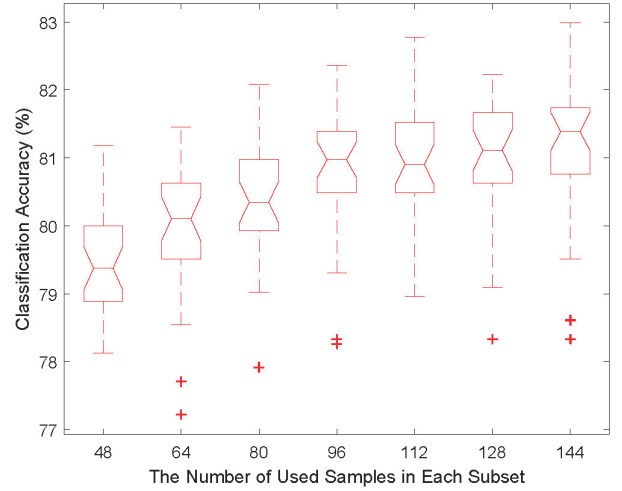


Fig. 2. Under different numbers of base learners, the impact of the number of samples in each subset on classification accuracy. In the boxplot, each box represents the distribution of classification accuracy when the number of base learners is 4, 12, 20, 40, 60, 80, 100, 120, 140 and 160.

IV. RESULTS

A. Influence of the number of samples in each subset on classification accuracy

In the TFDWEL method, each training subset obtains k samples from the original training samples through the bootstrap sampling method. The number of samples in the subset is related to the quality of the base learner model. In order to find the most suitable number of samples, we compared the impact of the number of samples in the subset on the classification accuracy under different numbers of base learners, as shown in Fig. 2. With different numbers of base learners, as the number of samples in the subset increases, the classification accuracy gradually increases. We can see that when the number of samples in the subset is 144, the median value and upper quartile (the top 25% results) of classification accuracy are higher than the other number of samples, so we choose $k = 144$ as the best number of subsets.

B. Influence of the number of base learners on classification accuracy

The number of component classifiers in ensemble learning is called ensemble size or ensemble cardinality [16]. Ensemble size will have a great impact on calculation cost and prediction performance, so determining its size is an important issue. In order to find the most suitable number of base learners, we compare the impact of the number of base learners on classification accuracy and computational time cost, as shown in Figure 3. We evaluate the computational time cost of the algorithm on a desktop computer with a i7-6700 CPU, 8 GB RAM, and an environment of MATLAB 2019a.

From Fig. 3, as the number of base learners increases, the classification accuracy first rises steadily, and then tends to flatten out, but the computational time cost is always increasing. In addition, when the number of base learners is 36, it is exactly at the intersection of the rising stage and

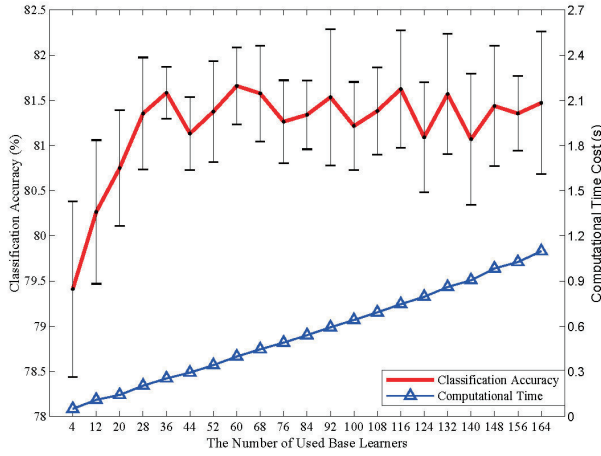


Fig. 3. The influence of the number of base learners on classification accuracy and computational cost. The red line represents the relationship between the number of base learners and the classification accuracy, and the black line represents the standard deviation of the classification accuracy of 9 subjects. The blue line represents the relationship between the number of base learners and the computational time cost.

the flat stage of classification accuracy. When increasing the number of base learners gradually, the classification accuracy increases very slowly, and may even decrease, and it will also increase the computational time cost greatly. Therefore, we choose 36 as the optimal number of base learners.

C. Analysis of the predictive ability of the ensemble learning model

A good ensemble classification model should have the characteristics of "accuracy" and "diversity", and training a good ensemble classification model is the prerequisite for obtaining efficient classification performance. We analyze the quality of the ensemble learning classification model trained by the TFDWEL algorithm. The prediction results of the base learner trained on the EEG data of subject 5 and subject 7 on the test set are shown in Fig. 4. It should be noted that the results in the Fig. 4 are obtained when the number of base learners is 12. The dark blue grid and blue-green grid in Fig. 4(a) and Fig. 4(c) indicate that the class predicted by the TFDWEL algorithm is correct, and the yellow grid indicates that the algorithm predicted class is wrong, and the blue-green and dark blue grids represent left-hand and right-hand MI tasks, respectively.

It can be observed from Fig. 4 that some test samples in the test set are difficult to distinguish correctly, such as trial 11 to 14 in Fig. 4(a) and trial 1, 4, 6 and 12 in Fig. 4(c). Since the trained base learner group is very powerful, the base learner 3, 4, 8 and 11 in Fig. 4(a) and the base learner 4, 8 and 12 in Fig. 4(b) can distinguish the above-mentioned difficult test samples. This shows that the base classifier trained by the algorithm has good accuracy and diversity.

D. Analysis of decision fusion

In ensemble learning, a suitable fusion method allows the algorithm to obtain a better global prediction model. We

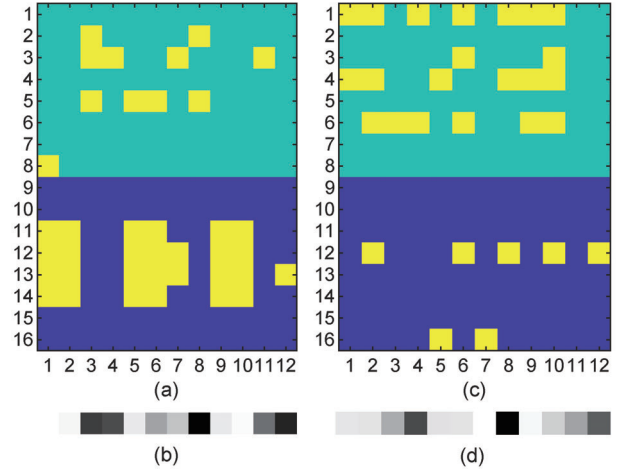


Fig. 4. Comparison of the predicted class of the base learner and the true class of the EEG test dataset, and the weight of each base learner. The abscissa in the figure represents the serial numbers of the 12 base learners, and the ordinate represents the sample serial numbers of the test set. Figures (a) and (b) show the results of subject 5, and Figures (c) and (d) show the results of subject 7. Figures (a) and (c) show the comparison between the predicted class of the base learner trained by the algorithm and the true class. Figures (b) and (d) show the weight of each base learner estimated by the least square error estimation algorithm. It should be noted that we normalize the weight value to the range from 0 to 1. The darker the grid color, the larger the weight value is, and vice versa.

further analyze the weights learned by the least square error estimation method. The weight values of the base learners of subject 5 and subject 7 are shown in Fig. 4(b) and Fig. 4(d), respectively. In the learning process of the least square error estimation method, the base learner with greater contribution to the final decision will be assigned a higher weight value, and the base learner with more misclassified samples will get a lower weight value. We can observe that the base learner 3, 4, 8 and 12 in Fig. 4(b) and the base learner 4, 8 and 12 in Fig. 4(d) are assigned higher weights. These base learners can accurately distinguish most test samples, which shows that the least square error estimation method can effectively assign appropriate base learner weights.

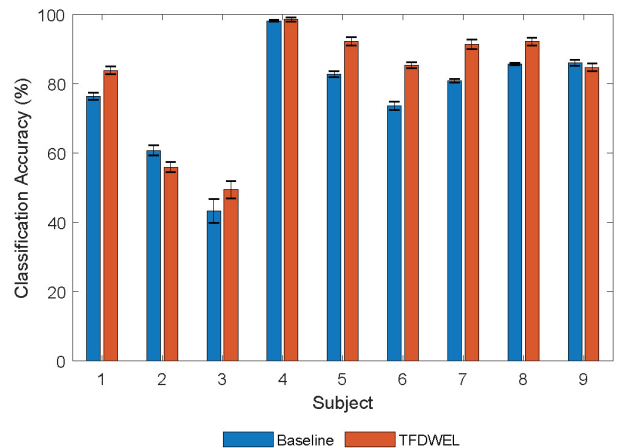


Fig. 5. The classification accuracy of 9 subjects obtained by Baseline method and TFDWEL method.

E. Comparison of classification performance

We compare the classification performance of the baseline method and the TFDWEL method, as shown in Fig. 5. In the baseline method, CSP extracts the features of EEG data with a frequency band of 8-35 Hz and a time window of 0.5-3.5 s . In the TFDWEL method, the optimal parameters are selected based on the above experiment, the number of subset samples is 144, and the number of base learners is 36. The classification accuracy of our proposed algorithm can reach 81.58%, which is 5.19% higher than the baseline method. Superior classification performance shows that the TFDWEL method is an effective solution to improve the classification performance of MI EEG signals.

V. CONCLUSIONS

In this paper, we propose a novel TFDWEL method to improve the classification accuracy of MI EEG signals. This method samples the EEG dataset to obtain multiple subsets with distribution differences, and obtains multiple time-frequency subsets with time-frequency diversity through time-frequency decomposition. Then the feature extraction model and the classifier model are trained to construct the base classifier, and accurately distinguish the MI task class by the weighted summation method. The TFDWEL method is evaluated on a public MI EEG signal dataset, and the excellent classification performance proves the effectiveness of the method in MI BCI.

In future work, we can optimize the base learner, try to use different types of classifiers and improve the fusion method of base classifier, and then obtain a powerful ensemble learning model to improve the classification accuracy of EEG signals.

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